

ADR-001: Megatrend Property & Lease Intelligence Platform

Status: Proposed — vendor-neutral draft **Date:** 2026-06-05 **Deciders:** Megatrend leadership, Cyphersoft (Andrew), AI/engineering lead [TBD] **Author:** Cyphersoft (Andrew) **Companion documents:** - Megatrend-Platform-Workflows.pdf — visual workflow reference - Megatrend-Platform-Overview.pdf — plain-English business overview

How to read this document

This ADR is intentionally vendor-neutral and cost-neutral. It captures **what the platform must do**, **the architectural decisions that need to be made**, and **the options worth weighing** — but it does **not** lock in specific products, vendors, hosting regions, or pricing.

The technology and cost columns are deliberately left blank, marked [TBD]. They are to be filled in by Andrew together with an AI/engineering lead during a shaping phase, once the principles in this ADR are agreed.

Where this document says "**the platform**" or "**the system**", it refers to a behaviour the solution must deliver, not a specific tool.

Table of Contents

1. Context
 2. Goals & Non-Goals
 3. Architectural Principles
 4. System Components (capability view)
 5. Decisions to be Made
 6. Non-Functional Requirements
 7. Workflows (summary)
 8. Data & Compliance
 9. Phasing
 10. Risks & Mitigations
 11. Action Items
 12. Open Questions
 13. Appendix A — Decision Worksheet (fill-in)
 14. Appendix B — Glossary
-

1. Context

Megatrend operates as a commercial property and lease management partner. Today's workflow is heavily manual — documents arrive over email, cataloguing happens in spreadsheets, and renewal sourcing relies on account-manager memory.

The proposed platform is a **dual-sided multi-tenant SaaS application**: a **Client Portal** for tenant-side users and a **Megatrend CRM Portal** for account managers, backed by an **AI analysis engine** and an **automated property-matching capability**.

1.1 Forces & constraints

Force	Description
Market	South African commercial property; POPIA compliance applies.
Scale (Y1, target)	[TBD — confirm during shaping] clients · [TBD] leases · [TBD] documents/month · [TBD] account managers.
Time-to-market	MVP target [TBD weeks]; confirm during shaping.
Team	[TBD — confirm engineering capacity].
Sensitivity	Commercially sensitive financial data. Immutable audit trail non-negotiable.
Budget	[TBD — to be set by Megatrend leadership].

2. Goals & Non-Goals

2.1 Goals

1. Eliminate the email-as-CRM workflow for client document intake.
2. Compress the time between a document arriving and an actionable insight from days to minutes.
3. Catch every approaching lease expiry early enough to source alternatives proactively.
4. Build a structured longitudinal record of every client portfolio (a long-term data moat).
5. Give Megatrend account managers a credible workspace to handle more clients without sacrificing quality.

2.2 Non-Goals (for this ADR / MVP)

- Detailed UI/UX design — handled in a separate design brief.
 - Pricing and packaging of the service to clients.
 - Customer-success operating model.
 - Replacing Megatrend's existing accounting or invoicing systems.
-

3. Architectural Principles

The principles below are technology-independent and should guide all subsequent decisions. They are the things we do **not** want to compromise on, even if vendor specifics change.

#	Principle	Implication
P1	Multi-tenant by construction	Tenant isolation is enforced at the data layer, not in application code alone. No design that requires the application to "remember" to filter by tenant is acceptable.
P2	POPIA-aligned data residency	Personal information stored in-region (South Africa) by default. Any cross-border processing must be specifically justified and contractually covered.
P3	Human-in-the-loop AI	The AI never publishes to a client without account-manager review. AI is a force multiplier, not an autonomous publisher.
P4	Async pipelines, responsive UIs	Heavy work (OCR, AI, scraping) is background. User-facing surfaces stay snappy and receive updates via real-time events.
P5	Tamper-evident audit trail	Every meaningful action is logged immutably. Records reference the hash of the previous record so any modification is detectable.
P6	Pluggable infrastructure	No vendor lock-in at the architecture level — provider-specific calls are abstracted behind internal interfaces so a vendor can be swapped without rewriting business logic.
P7	Compliance-first scraping	Partner APIs preferred. Any scraping is rate-limited, robots.txt-respecting, identified in User-Agent, and has a per-source kill switch. Legal review precedes each new source going live.
P8	Observability from day one	Structured logs, metrics, and traces in the MVP — not a Phase 2 retrofit.
P9	Idempotency & retries	Every pipeline step is safe to re-run. Failures retry; persistent failures land in a dead-letter queue and alert the on-call engineer.
P10	Sensible defaults, configurable per tenant	Lease-expiry horizons, notification preferences, and similar parameters can be tuned per client without code changes.

4. System Components (capability view)

This is a capability-level description — what each part must do — without naming products.

4.1 Client Portal

- Multi-tenant web experience accessible to client users.
- Authenticated access scoped to their own tenant only.
- Document upload (single and bulk; desktop and mobile-camera capture).
- Dashboard with portfolio summary, leases, key dates, and curated insights.
- Notification of meaningful events (new insight published, renewal options ready).

- Roles within the client organisation (e.g. admin, finance, property manager).

Form-factor decision (D1): Web app · Progressive Web App · Native mobile · Combination — see §5.

4.2 Megatrend CRM Portal

- Internal workspace for account managers and Megatrend admins.
- Inbox of new client uploads and AI-drafted insights, ranked by urgency.
- Property and lease record management.
- Searchable document repository (full-text and semantic).
- Curation interface to edit, approve, reject, and publish insights to a client dashboard.
- Tenant administration (create client, assign team, configure thresholds).
- Audit log explorer for any tenant the user is authorised to view.

4.3 Document Ingestion & OCR Capability

- Accept any document the client uploads (PDF, image, scanned).
- Validate (file type, size, malware).
- Extract structured data (key-value pairs, tables) from invoices, utility statements, lease agreements.
- Surface low-confidence extractions for human review rather than silent failures.

OCR strategy decision (D2): managed cloud OCR · self-hosted OCR · LLM vision · combination — see §5.

4.4 AI Analysis Engine

- Classify incoming documents (invoice, utility, lease, statement, other).
- Extract structured data; normalise vendor names, units, tariff codes.
- Detect anomalies in spending patterns and utility usage.
- Forecast utility costs and other recurring spend.
- Generate plain-language narrative insights with linked source documents and a self-reported confidence score.
- Improve over time from account-manager edits and rejections.

Model strategy decision (D3): which LLM(s), which forecasting approach, which embedding model — see §5.

4.5 Property Scout

- Watch every lease across every tenant; alert when within a configurable renewal horizon.
- Source comparable properties via official partner APIs where available.

- Fall back to compliant scraping of public listings only when APIs are insufficient.
- Normalise listings to a common schema, deduplicate, score against lease criteria.
- Produce a comparative business-case report for account manager review.

Data source decision (D4): which partner APIs to license; which sources to scrape — see §5.

4.6 Workflow Orchestration

- Coordinate the document upload pipeline (validate → classify → extract → index → notify → analyse).
- Coordinate the property-scout pipeline (watch → match → scrape → score → report → notify).
- Provide retries, dead-letter handling, scheduled triggers, and observability for both.

Orchestration tooling decision (D5): managed workflow engine · self-hosted workflow engine · custom code — see §5.

4.7 Real-time Event Bus

- Tenant-scoped channels for live updates between client and CRM portals.
- JWT-secured; each tenant can only subscribe to its own events.
- Survives reconnection; degrades gracefully to polling if the connection is lost.

Real-time delivery decision (D6): managed service · self-hosted · other — see §5.

4.8 Identity & Tenancy

- Multi-tenant identity with organisations, roles, and (for enterprise clients) SSO support.
- MFA mandatory for Megatrend staff; optional for client users.
- Tenant ID resolved from the authenticated session and propagated to every downstream call.

Identity provider decision (D7): managed identity service · self-hosted · build · combination — see §5.

4.9 Data Layer

- Primary database for structured records (tenants, properties, leases, documents metadata, insights, audit).
- Object storage for original document files.
- Index for full-text and semantic search.
- Cache and queueing for the async pipelines.

Database & storage decisions (D8): see §5.

4.10 Audit & Compliance

- Tamper-evident log of every meaningful action (client, Megatrend, system, admin, infrastructure).
- Mirror to a write-once archive with long retention.
- Queryable from the CRM for day-to-day inspection.
- Exportable per tenant for client transparency and POPIA requests.

4.11 Observability & Operations

- Structured logs, metrics, and traces from every component.
- Alerting on error rates, latency anomalies, and failed jobs.
- Runbooks for the on-call engineer.

Observability stack decision (D9): see §5.

4.12 Hosting & Infrastructure

- In-region hosting (South Africa) for tenant data.
- Infrastructure-as-code so environments can be recreated and audited.
- Separate environments for development, staging, production, and audit logs.

Hosting decision (D10): which cloud / region / infrastructure model — see §5.

5. Decisions to be Made

Each decision below has options, key trade-offs, and a blank choice field for Andrew + the AI/ engineering lead to fill in during the shaping phase.

D1 — Client portal form factor

Option	Optimises for	Trades against
Responsive web only	Speed to ship, lowest cost	No offline upload; no native camera UX
Progressive Web App	Single codebase + mobile install + offline	iOS push notification limits
Native iOS + Android + web	Best mobile UX, full background sync	Three codebases; higher build + maintenance

Choice: Rationale:

D2 — OCR strategy

Option	Optimises for	Trades against
Managed cloud OCR	Accuracy on invoices/tables; zero ops	Per-page cost; vendor dependency
Self-hosted open-source OCR	Cost at scale; full control	Lower accuracy on complex layouts; ops overhead
LLM vision only	Handles unstructured/odd layouts; one fewer vendor	Higher per-document cost; less deterministic
Hybrid (structured OCR + LLM fallback)	Best accuracy/cost balance	More moving parts

Choice: Rationale:

D3 — AI model selection

Sub-decisions inside this single decision: classifier model, extraction model, embedding model, forecasting approach.

Lever	Considerations
Classifier	High-volume, low-cost path. Lean toward smaller, faster models.
Extraction	Higher-stakes accuracy. May warrant a larger model + structured outputs.
Embeddings	Cost per million tokens; in-region API availability; quality on financial/legal text.
Forecasting	LLMs are unreliable for numeric forecasts. Classical time-series methods are usually cheaper, more interpretable, more accurate.

Choice: Rationale:

D4 — Property data sources

Lever	Considerations
Partner APIs	Higher data quality; licensing cost; legally clean.
Compliant scraping	No licence fee; legal/ToS risk; brittleness; per-source kill switch needed.
Geographic coverage	Confirm which SA regions and property types matter most for current client portfolios.

Choice (sources to license): Choice (sources to scrape, with legal sign-off):

D5 — Workflow orchestration

Option	Optimises for	Trades against
Managed serverless workflow engine	Zero ops; native retries; great DX for code-first workflows	Per-execution cost; vendor lock-in
Self-hosted workflow engine (visual or code-first)	Cost at high volumes; full control	Ops burden; needs HA design
Pay-per-task automation platform	Fast to wire	Cost explodes with volume; weak observability
Custom code + queue library	Full control; minimal dependencies	Reinvents retries/scheduling; engineering tax

Choice (MVP): Choice (Phase 2 critical paths):

D6 — Real-time delivery

Option	Optimises for	Trades against
Managed pub/sub service	Zero ops; tenant ACLs out of the box	Per-connection cost
Self-hosted websocket + cache adapter	Cheap at scale; in-region	Ops burden; sticky sessions; reconnection logic

Choice: Rationale:

D7 — Identity & SSO

Option	Optimises for	Trades against
Managed B2B identity provider	Fast to ship; org/role support; SSO-ready	Per-MAU cost; vendor lock-in
Self-hosted open-source identity	No licence fee; full control	Ops burden; security responsibility
Build on top of cloud-native identity primitives	Fits if all infra is on one cloud	More glue code

Choice: Rationale:

D8 — Database, object storage, search

Lever	Considerations
Primary DB	Must support row-level tenant isolation, JSON, full-text + vector search.
Object storage	In-region; encryption with customer-managed keys; write-once option for audit copies.
Search	Built into the primary DB vs. a dedicated search engine — depends on volume and query patterns.

Choice (DB): Choice (Object storage):
Choice (Search):

D9 — Observability

Lever	Considerations
Errors	Aggregate exceptions across services; tie to releases.
Logs / uptime	Centralised log search; uptime probes; alerting.
Traces & metrics	Distributed tracing across the async pipeline; SLO dashboards.

Choice: Rationale:

D10 — Hosting & infrastructure

Lever	Considerations
Cloud provider	Must have in-region capacity for SA; compliance posture; team familiarity.
Region(s)	Primary in SA; consider an out-of-region backup for audit copies only.
IaC	Whatever the team is fastest in; required for production.
Environments	At minimum: dev, staging, production, audit.

Choice: Rationale:

D11 — Multi-tenancy isolation model

Option	Isolation	Cost	Operability
Silo (database per tenant)	Strongest	High	Hardest
Bridge (schema per tenant)	Strong	Medium	Medium
Pool (shared schema with tenant ID + row-level security)	Software-enforced	Low	Easiest

Choice: Rationale:

Note: any enterprise client that demands a silo deployment will warrant an architectural exception captured as its own ADR.

6. Non-Functional Requirements

Targets to set during shaping; defaults shown for the team to confirm or revise.

Dimension	Target	To confirm
Availability	99.5% MVP, 99.9% Phase 2	✓
Upload acknowledgement	< 2s	✓
Insight publication latency (p95)	< 5 min	✓
Search latency (p95)	< 400ms over [TBD] documents	✓
Security	TLS in transit; encryption at rest with managed keys; MFA for staff	✓
POPIA — DSAR	Fulfilled in 30 days	✓
Auditability	All actions immutably logged, 7-year retention	✓
Cost	[TBD]	✓

7. Workflows (summary)

The companion document **Megatrend-Platform-Workflows.pdf** contains an annotated diagram and narrative for each of these six workflows. Captured here only by name:

1. **Client document upload & AI pipeline** — ingest, classify, extract, index, notify, analyse, surface.

2. **Lease expiry watcher & property scout** — scheduled scan → match → score → business-case report.
 3. **AI insight curation** — AI drafts → account-manager review → publish.
 4. **New client onboarding** — wizard-driven tenant creation, isolation, team assignment.
 5. **Real-time sync between portals** — tenant-scoped channels keep both sides live.
 6. **Audit & compliance trail** — tamper-evident chain mirrored to a write-once archive.
-

8. Data & Compliance

8.1 Data the platform holds

- Client identity (organisation name, contacts, users)
- Property records (address, type, metadata)
- Lease records (terms, dates, rates, escalation, landlord)
- Document files (original uploads) + extracted structured data
- AI insights (draft, curated, published)
- Audit events (all of the above + system / admin activity)

8.2 POPIA controls

Control	Description
Lawful basis	Captured in client agreement at onboarding.
Data residency	South Africa primary; any cross-border processing covered by processor agreements.
Encryption	At rest (customer-managed keys) and in transit (TLS).
Access	RBAC + tenant isolation enforced at the data layer.
Retention	Defined per data class; documented in a Retention Schedule.
Data-subject rights	Export and deletion supported within 30 days.
Breach response	Documented playbook; regulator notification process.
Sub-processors	Each vendor signed under a processor agreement; list maintained and disclosed.

9. Phasing

Phase 1 — MVP (target: [TBD weeks])

Goal: a working platform both sides log into every day.

- [] Tenant infrastructure stood up (hosting, identity, storage, observability).
- [] Client Portal: auth, dashboard, document upload, basic insight view.

- ☐ Megatrend CRM: inbox, lease/property CRUD, search, manual insight publishing.
- ☐ Document ingestion + extraction pipeline.
- ☐ Real-time notifications.
- ☐ Tamper-evident audit log.
- ☐ Two or three pilot clients onboarded.

Phase 2 — Intelligence & automation (target: [TBD weeks after MVP])

Goal: the platform starts working on its own.

- ☐ Property Scout (lease expiry watcher + matching engine).
- ☐ Automated insight generation (anomalies, forecasts).
- ☐ PWA/mobile polish.
- ☐ Optional CRM integrations (HubSpot / Salesforce / equivalents).
- ☐ Optional payment integration.

Phase 3 — Scale & ecosystem (timing TBD)

- ☐ Native mobile apps (if usage justifies).
 - ☐ White-label offering.
 - ☐ Benchmarking marketplace.
 - ☐ Public API for third parties.
-

10. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Scraper triggers ToS / legal complaint	M	H	Partner APIs first; documented compliance policy; legal review per source; per-source kill switch.
Cross-tenant data leak	L	Severe	Data-layer isolation; quarterly cross-tenant penetration test; documented incident response.
Vendor outage in a critical path	L	M	Async architecture absorbs short outages; fall-back paths designed in (e.g. structured OCR ↔ LLM vision).
AI hallucination on financial extraction	M	H	Confidence scoring; mandatory account-manager review before client visibility; source document always one click away.
Single point of failure in orchestrator	M	M	High-availability deployment; durable queue; migrate critical paths to a durable workflow engine in Phase 2.
Vendor lock-in	M	L	Provider-specific code behind internal interfaces; documented exit plan for each vendor.
Scope creep (CRM becomes a generic CRM)	H	M	Roadmap discipline; integrate with existing CRMs in Phase 2 rather than competing with them.
Budget overrun	M	M	Cost ceiling per environment; usage-based services capped; monthly cost review.

11. Action Items

1. [] Shaping session: Andrew + AI/engineering lead to fill in §5 (D1–D11) and §6 NFR targets.
2. [] Engage legal counsel for POPIA processor agreements with eventual vendors.
3. [] Identify and approach property data partners.
4. [] Confirm hosting environment(s) and provision base infrastructure-as-code repo.
5. [] Recruit 2–3 pilot clients for Phase 1.
6. [] Define cost ceiling and per-month review cadence.
7. [] Schedule internal ADR review within 10 working days; sign-off gates Phase 1 kickoff.

12. Open Questions

- What is Megatrend's current system of record for leases, and how do existing records migrate in?
- Any enterprise clients who would require a private/silo deployment?
- Which 2–3 pilot clients are the target for MVP rollout?
- Is there an existing identity provider (e.g. Microsoft 365, Google Workspace) that Megatrend staff already use?

- Are there any commercial relationships or constraints that pre-determine any of the D1–D11 decisions?
- What is the budget envelope — both for Phase 1 build and for monthly operating cost?

Appendix A — Decision Worksheet (fill-in)

A single-page template you can print, fill in during the shaping session, and circulate back into this ADR.

#	Decision	Options considered	Choice	Rationale	Decision date	Decider(s)
D1	Client portal form factor	Web · PWA · Native · Combination				
D2	OCR strategy	Managed · Self-hosted · LLM · Hybrid				
D3	AI models	Classifier · Extraction · Embeddings · Forecasting				
D4	Property data sources	Partner APIs · Compliant scraping				
D5	Workflow orchestration (MVP)	Managed serverless · Self-hosted engine · Pay-per-task · Custom				
D5b	Workflow orchestration (Phase 2)	(as above)				
D6	Real-time delivery	Managed pub/sub · Self-hosted				
D7	Identity & SSO	Managed · Self-hosted · Cloud-native primitives				
D8	Database, storage, search	(per sub-lever)				
D9	Observability	Errors · Logs · Traces/metrics				
D10	Hosting & infrastructure	Cloud · Region · IaC · Environments				
D11	Multi-tenancy model	Silo · Bridge · Pool				

Item	Target / value
Y1 client count	
Y1 leases	
Y1 documents/month	
MVP duration (weeks)	
Phase 2 duration (weeks)	
Phase 1 build budget	
Monthly operating budget (Y1)	
Property data licensing budget (Y1)	
Availability target (MVP / P2)	
Insight publication SLA (p95)	

Appendix B — Glossary

Term	Meaning
ADR	Architecture Decision Record
AM	Account Manager (Megatrend internal user)
DSAR	Data Subject Access Request (POPIA)
MFA	Multi-Factor Authentication
NFR	Non-Functional Requirement
OCR	Optical Character Recognition
POPIA	Protection of Personal Information Act (South Africa)
PWA	Progressive Web App
RBAC	Role-Based Access Control
SSO	Single Sign-On
TBD	To Be Decided — pending shaping session
WORM	Write Once Read Many (immutable storage)